IAEA Coordinated Research Project on Development of Harmonized QA/QC Procedures for Maintenance and Repair of Nuclear Instruments

Test Procedure for PIN Diode Radiation Detectors

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CONTENT

PAGE

1.- OBJECTIVE AND SCOPE 3

1.1.- Objective 3
1.2.- Scope 3

2.- NOTATION AND DEFINITIONS 3

2.1.- Notation 3
2.2.- Definitions 3

3.- DEVELOPMENT 4

3.1.- Generalities 4
3.2.- Introduction 4
3.3.- Test Instruments 5
3.4.- Test Conditions 5
3.5 .- Test Circuits 7
3.6.- Measurements 8
3.7.- Detector Aging or Damage 12

4.- ADMINISTRATION OF THE TEST REPORTS 14

4.1.- Numbering of the reports 14
4.2.- Personnel 14
4.3.- Test Report 14

5.- ACTION IN CASE OF NON CONFORMITIES 14

5.1.- Technical Report 14
5.2.- Labelling 14

6.- RESPONSIBILITIES 14

6.1.- Head of the Department 14
6.2.- Area Responsible 14
6.3.- Operative personnel 15

7.- BIBLIOGRAPHY 15

8.- ANNEXES 15

1.- OBJECTIVE AND SCOPE

1.1.- Objective

The objective of this procedure is to describe the steps to verify the radiation response and the electrical characteristics of PIN diode semiconductor detectors employed in counting and spectroscopy of X-rays and charged particles.

1.2.- Scope

This procedure is applicable only for the verification and diagnostic of small area PIN photodiodes used as radiation detectors. Some low cost devices that can be used are: OPF420, S3883, HP 5082-4207.

2.- NOTATION AND DEFINITIONS

2.1.- Notation

PIN	P type region, Intrinsic region, N type region structure of a special diode.
MCA	Multichannel Analyzer
DMM	Digital multimeter

2.2.- Definitions

2.2.1. Detector
A device that converts the energy of a photon or incident particle in an electric pulse.

2.2.2. PIN diode
A diode made of silicon with a structure formed by a p-type layer, an intrinsic region with high resistivity and a n-type layer.

2.2.3. Dead Layer
A region in the detectors in which no useful ionization is produced.

2.2.4. Detector Background Counting
Counting that is present even without a radiation source, when the detector is biased.

2.2.5. Resolution
Capability of a radiation detector to distinguish between two peaks of next energies. For PIN diodes, it is expressed in keV´s.

2.2.6. Charged particles
The charged particles considered in this procedure are: electrons, protons, alpha and beta particles and nuclei of light and heavy elements.

3.- DEVELOPMENT.

The sequence of steps to verify the radiation response and electrical characteristics of PIN diode detectors is described in the next paragraphs. A flux diagram of the process is shown in Annex I.

3.1- Generalities

A technical report of the verification of the electrical characteristics and the response to radiation of PIN diode detectors must include the circuit diagram, environmental conditions, geometry of the testing set-up, count rate, isotope and test instruments employed.

3.2- Introduction

PIN photodiodes have a great versatility as radiation detectors because can be used both for X-ray measurements and also for charged particles. Some interesting applications could be generated having as radiation detector this kind of devices. A PIN diode spectrometer consist of a radiation detector, signal processing electronics interfaced to a pulse height analyzer and a computer, see Fig.1.

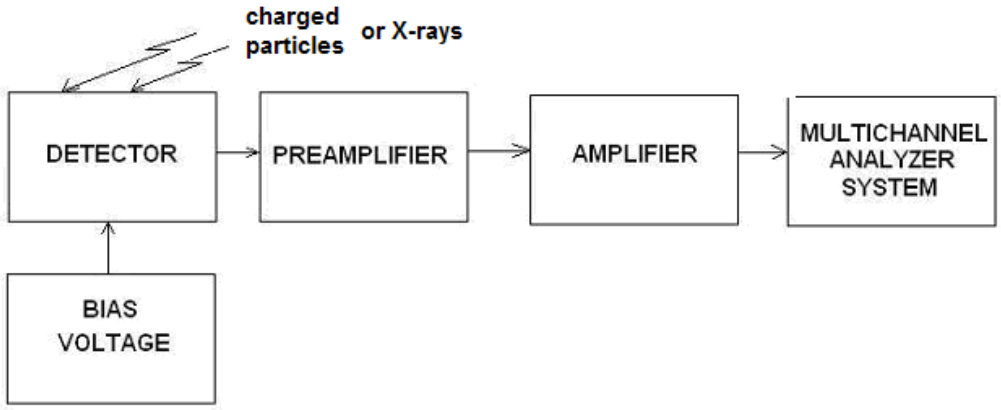


Fig. 1.- Blocks diagram of a PIN diode spectrometer

When a potential difference is established between the electrodes of a PIN diode, an electric field in produced in the semiconductor. A charged particle has a very great ionisation capability, when enters the semiconductor, it interacts and produces free charge carriers in the crystal, the number of which is proportional to the energy lost by the particle, a similar result is obtained when X-rays interact with the detector. The charge motion resulting from the influence of the electric field produces an induced current pulse in the external circuit. The integrated current pulse is proportional to the energy lost by the charged particles or the energy delivered by X-ray photons.

The pulses are routed to a MCA where they are sorted and stored according to the amplitude distribution to produce a pulse-height graph that corresponds to the energy spectrum of the charged particles or X-rays. The MCA may be a dedicated instrument or an analog-to-digital converter (ADC) interfaced to a computer.

PIN diodes are made with high resistivity silicon to withstand large electric fields with very low leakage currents, for better performance for X-rays detection, the leakage current must be less than 100 pA at room temperature.
Energy resolution is one of the most important characteristics of an energy spectrometer since it sets the limit on the ability to resolve closely spaced lines. Other important parameters are count-rate capability and detector size (efficiency).

Effects of high count rate such as pulse pile-up and dc level shifts should be considered. Every time an event produces a pulse in the amplifying equipment, the dc levels throughout the system are perturbed and take some time to return to their original values. If another event occurs within this time interval, its effective output pulse height may be altered, thereby contributing to spectral distortion.

3.3.- Test Instruments

All the instruments employed in the tests must be calibrated and with a valid calibration certificate.

3.3.1 Detector Bias Power Supply.
Generally it is a voltage power supply with low current capability to feed the detector, regulation less than 0.1 % is desired. The ripple should be less than 100 mV.

3.3.2 Pulse Amplifier.
A spectroscopy amplifier with shaping times less than 3 μs is required.

3.3.3 Oscilloscope
Use an analog or digital oscilloscope.

3.3.4. Multichannel Analizer, MCA
A MCA with at least 1024 channels is needed to get the energy spectrum. Consider that the analog input pulse must be positive.

3.3.5. Electrometer
The measurement of very low currents can be made with an electrometer.

3.4.- Test Conditions

3.4.1.- Background Radiation
Be sure that the only contribution to the detector counting is the natural background, avoiding the contribution due to any additional radioactive source.

3.4.2.- Temperature
The characteristics of PIN diode detectors are temperature dependant, a reference temperature between 20º C and 25° C is considered, the operation at lower temperatures

reduces the thermal noise of the detector and improves the resolution obtained in the spectrometer.

3.4.3.- Light leakage

PIN diode detectors are very sensitive to the visible light, then any light leakage must be avoided during the tests.

3.4.4.- Vacuum conditions

In the measurement of charged particles, the air interacts with the particles, reducing its apparent energy, in order to get the best results, a good vacuum, better than 1×10^{-2} Pa (1×10^{-4} mbar), must be established in the measuring chamber.

3.4.5.- Radiation flux

PIN diode detectors can not operate at very high radiation fluxes because they has a limitation related with the number of incident particles per unit of time and its amount of energy, in normal applications the radiation flux is less than 100 000 MeV/s.

3.4.6.- Radioactive source employed for a charged particle spectrometer.

Generally, an Am-241 alpha source is specified to characterize the PIN diodes for charged particles.

The energy calibration of the spectroscopy system can be done with a triple source containing Am-241, Pu-239 and Cm-244. See Table 1, for details. Sources with an activity of around 37 000 Bq (1 μCi) could be used.

Isotope	Radiation	Energy (MeV) (% yield)	Mean life
^{239}Pu (Plutonium)	alpha	5.104 (11.5 %) 5.142 (15.1 %) **5.155 (73.3 %)**	24 000 years
^{241}Am (Americium)	alpha	5.389 (1.3 %) 5.443 (12.7 %) **5.486 (86 %)**	458 years
^{244} Cm (Curium)	alpha	5.762 (23.6 %) **5.804 (76.4 %)**	18 years

Table 1. Radioactive sources used for energy calibration.

3.4.7.- Radioactive source employed for an X-ray spectrometer.

Generally, an Am-241 photon source is specified to characterize the PIN diodes in an X-ray spectrometer. Sources with an activity of around 37 000 Bq (1 μCi) could be used. The characteristics of the source are in the Table 2.

Isotope	Radiation	Energy (keV) (% de ocurrencia)	Mean life
^{241}Am (Americium)	X Rays	11.9 (0.86 %) 13.9 (13.2 %)	458 years
	Gamma Rays	17.8 (19.25 %) 20.8 (4.85 %) 26.35 (2.4 %) 59.54 (35.9 %)	

Table 2. Photon radioactive source used for energy calibration.

3.5. Test Circuits

To assemble a spectrometer with a PIN diode, a special preamplifier is required, because the detector has a very low capacitance and low noise is required, also, the connection between detector and preamplifier must be as short as possible. The preamplifier shown in the Fig. 2 is employed, in order to use it for charged particles, the value of the feedback capacitor, Cf, is 1 pF. For X-rays its value is 0.045 pF.

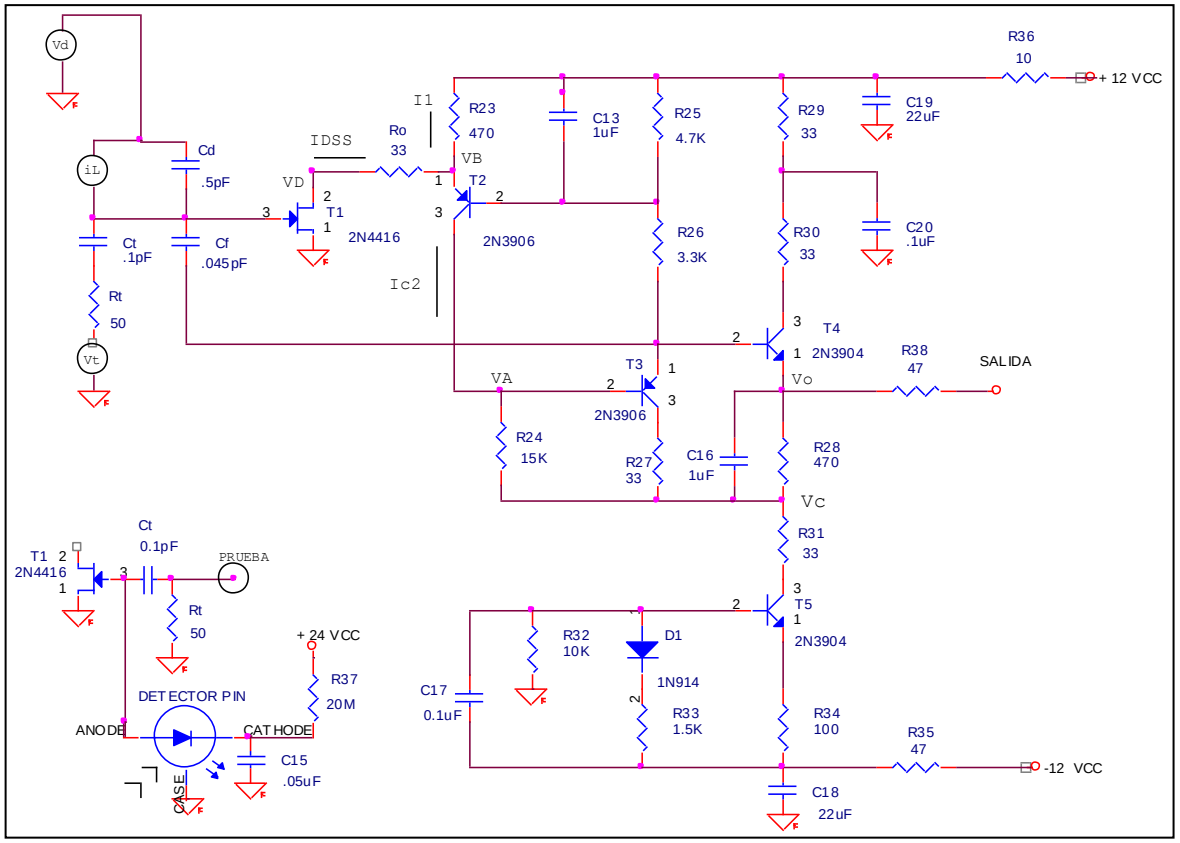


Fig. 2 .- Charge sensitive preamplifier suggested for small area PIN diodes.

3.6.- Measurements

In this paragraph, the measuring techniques employed to obtain the parameters of PIN diode detectors are defined.

3.6.1. Measurement of the rectifying junction.

The rectifying junction of the detector can be measured with a DMM, in the diode option, in forward bias condition, the reading should be around 0.6 V, also it could be measured with a curve tracer, see Fig. 3. In reverse bias condition, the behavior is almost like an open circuit, because the leakage current is very small (less than 1 nA).

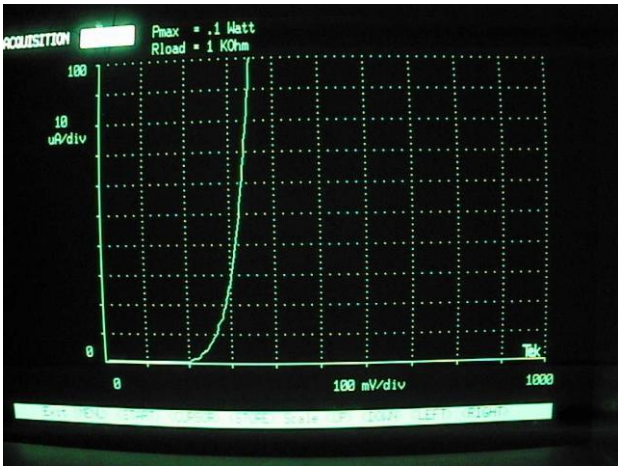


Fig. 3.- I-V curve for a PIN diode, the ranges in the curve tracer are: 10 μA/div., 100 mV/div.

3.6.2. Reverse current.

The reverse current in a PIN diode must be less than 100 pA, if a good energy resolution is required in X-ray spectroscopy. Fig. 4 shows the circuit used to make the measurement of reverse current. An electrometer is required (for example: Keithley 610C, Keithley 6517A) and a voltage power supply. If light is entering the detector surface, it contributes drastically to the increase in the reverse current.

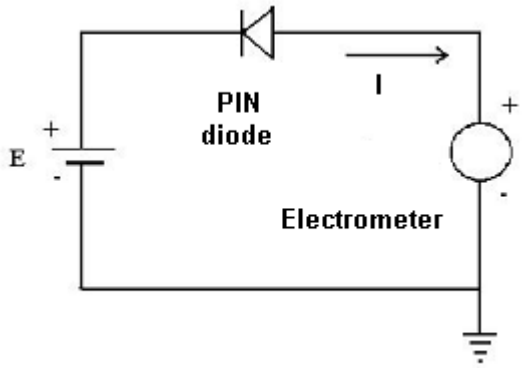
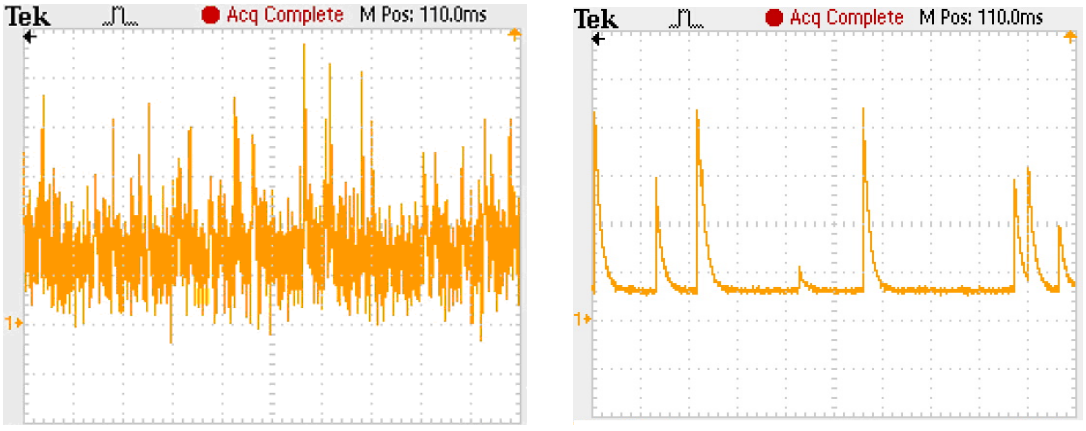


Fig. 4. Measurement of the leakage current of a PIN diode.

3.6.3. Energy Resolution *R*, of the detector in a charged particle spectrometer.

The PIN diode has a glass window, chaged particles could not pass easyly through the window, therefore, the window must be removed in this application.
Connect the detector as shown in Fig.1. Put the triple radioactive source, see pharagraph 3.4.6, inside the vaccum chamber, close to the detector, make vaccum to the measuring chamber, verify the correct polarity and value of the bias voltage according with the data sheet and apply the bias voltage, some pulses will appear in the oscilloscope screen at the preamplifier output, even without the bias voltage. See Fig. 5.a, the pulses could be positive or negative depending of the design of the preamplifier, notice that the noise reduces with the applied voltage, see Fig. 5.b.



a) b)

Fig. 5.- Pulses at the preamplifier output, a) without the bias voltage; b) with the bias voltage. The settings of the oscilloscope are: 0.5 V/div. and 100 μs/div.

Be sure that the counts be less than 5000 counts per second to avoid errors by pile up or dead time.

Adjust the shaping time of the amplifier to 2 μs or less and adjust the pole-zero for the best unipolar signal at the output of the amplifier, see Fig. 6.

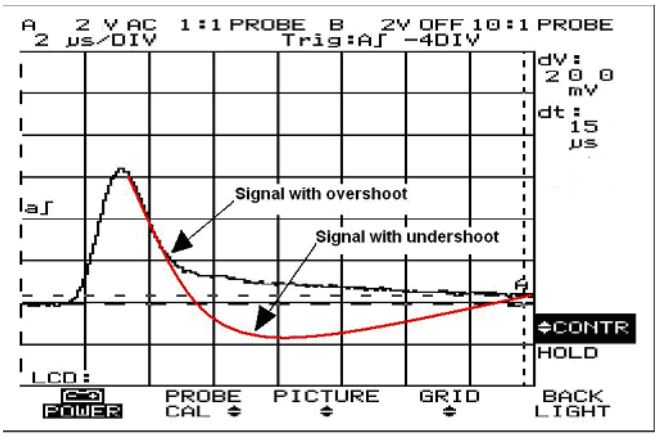


Fig. 6.- Amplifier output signal without proper compensation. The settings of the oscilloscope are: 200 mV/div. and 20 μs/div.

Adjust the amplifier gain to have a pulse of 6 V aproximately, at the output of the amplifier for the Am-241 peak.

Accumulate in the MCA to get more than 10000 counts in the peaks, to minimize the relative uncertainty of the results, < 1%.

Calibrate the MCA in energy units, by using the computer routine for that purpose or obtain the energy calibration equation for the Spectrometry System:

$$y = m\,x + b$$

where: y = energy value m = energy per channel

x = channel value b = energy at channel 0

Employ the three main peaks of the triple source for this porpuse, the energie value of the main peaks are marked in Table 1.

If you made the calibration process using the software routine, then obtain the resolution R, measuring the FWHM in the Am-241 peak, using the MCA.

On the other hand, if you made the calibration manually and you know the value of energy per channel, *m*, get the number of channels at the FWHM and then, convert them to an energy value, see Fig. 7 for the definition of:

$$R = FWHM$$

where:

FWHM = full width at half of maximum in keV

Ho = 5.486 MeV

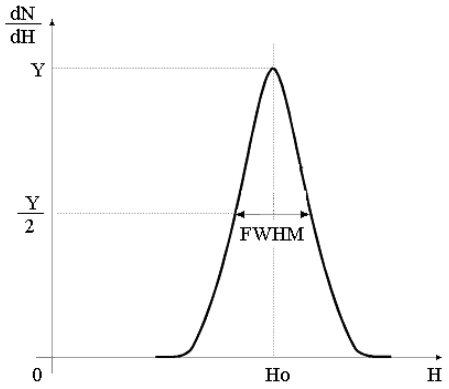


Fig. 7.- Definition of energy resolution.

In the Fig. 8, there is an example of the energy spectrum for a triple source obtained with a PIN diode detectors in a MCA with 1024 channels. The three peaks of Pu-239, Am-241 and Cm-244 are shadowed.

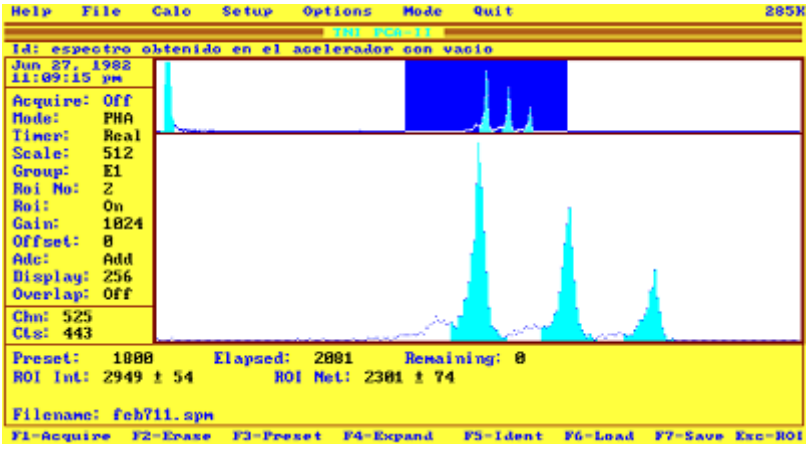


Fig. 8.- Energy spectrum of a triple source obtained with a PIN diode detector.

In this case, from Fig. 8, the resolution is :

$$R = 18 \text{ keV}$$

3.6.4. Energy Resolution *R*, of the detector in an X-ray spectrometer.

3.6.4.1 Use an ^{241}Am radioactive source which emits X-rays of 13.9 keV, 17.8 keV, and 20.8 keV and gamma rays of 26.35 keV and 59.54 keV. Place the source at an adequate distance to obtain a pulse repetition rate not greater than 2 000 cps to minimize dead time effects.

3.6.4.2 Acquire an energy spectrum for 10 minutes or the enough time to get a net count in the peaks of around 10000 counts to minimize the relative uncertainty of the results, < 1%. Use the peak energies of 13.9 keV and 17.8 keV to calibrate the MCA as shows in Fig. 9.

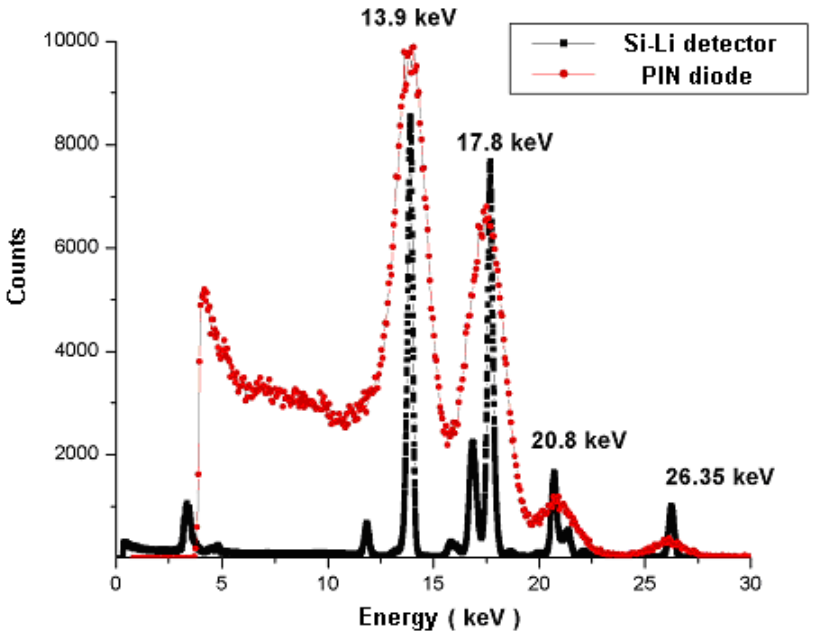


Fig. 9.- Energy spectrum of an Am-241 source obtained with a PIN diode detector.

3.6.4.3 Measure the FWHM in the peak of 13.9 keV of the Am-241 source, using the calibrated MCA.

3.6.4.4 The result must be around 1.8 keV, if you get a too much higher value, reduce very carefully the bias voltage (steps of 1 V every 5 seconds) until you have zero volts, turn off the low voltage power supply, dry carefully the preamplifier with warm air and check that all the BNC coaxial cables are in good conditions.

In the Fig. 9, also the spectrum obtained with a Si-Li detector is shown showing the great accuracy of the results with the PIN diode

3.7.- Detector Aging or damage.

The aging or damage of PIN detectors affects its detection characteristics and parameters. With respect to the functionality, the more important effects are:

- Reduction in the pulse size due to a dirt surface
- Increase of background counts due to radioactive contamination.

- Increase of the reverse leakage current, see Fig. 10, and of the threshold voltage in forward bias condition, Fig. 11, due to a damage produced by an excessive flux of charged particles,. The PIN diode was exposed in this case to a 12 nA beam of C-12 with an energy of 13 MeV.

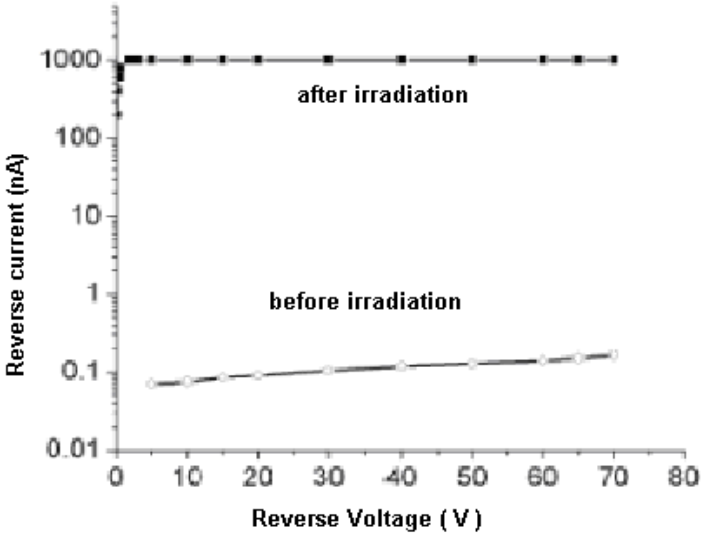


Fig. 10. Effect of irradiation damage in the reverse current of a PIN diode.

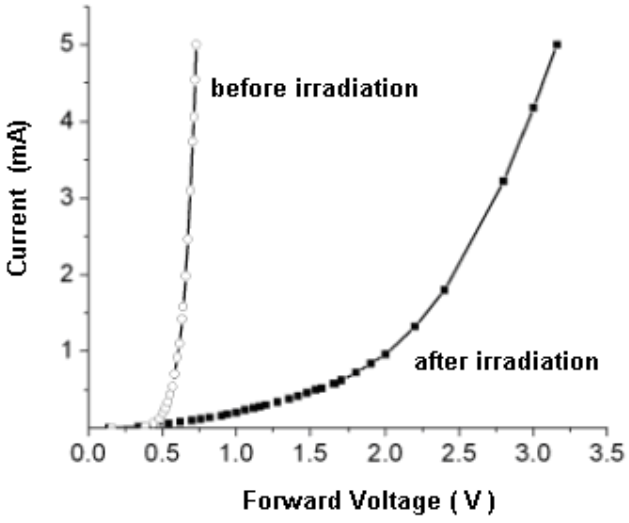


Fig. 11. Effect of irradiation damage in the threshold voltage in forward condition of a PIN diode.

4.- ADMINISTRATION OF THE TECHNICAL REPORTS

4.1 Numbering of the reports.

All the generated technical reports must have a unique and consecutive number.

4.2.- Personnel.

The test of PIN diode detectors must be done by trained personnel.

4.3.- Technical Report of results.

4.3.1 The results of the test of PIN diode detectors must be registered in a unique technical report, stating the description of the detector, mark, model, serial number, and all the test conditions, including the name of the person who made the tests.

4.3.2 All the technical reports must be classified and keep in a folder for future consult.

5.- ACTION IN CASE OF NON CONFORMITIES.

5.1 Technical Report.

Even in the case that results of the test are not as expected, a technical report has to be elaborated, indicating the non conformities and how far are the measured characteristics from the ideal ones.

5.2 Labelling.

The components or equipments that are not under specifications or with a failure have to be marked with a label indicating: **OUT OF SPECIFICATIONS** and **FAILURE** respectively.

6.- RESPONSIBILITIES

6.1.- Head of the Department.

Supervise that all the activities for testing of PIN diode detectors follow the established procedure.

6.2.- Area Responsible.

6.2.1 Assure that all the electronic test equipment be in good operational conditions.

6.3.2 Verify that all the activities for testing of PIN diode detectors follow the established procedure

6.2.3 Verify that the technical reports contain all the details of the testing of charged particle semiconductor detectors.

6.2.4 Maintain a register and control of the technical reports for all detectors tested in the laboratory.

6.3 Operative Personnel.

6.3.1 Verify that all the electronic test equipment be in good operational conditions.

6.3.2 Follow the steps established in this procedure for the testing of PIN diode detectors.

6.3.3 Elaborate the technical report of all the tests of the detectors.

6.3.4 Inform to the Area Responsible of any anomalous condition encountered during the test procedure.

7.- BIBLIOGRAPHY

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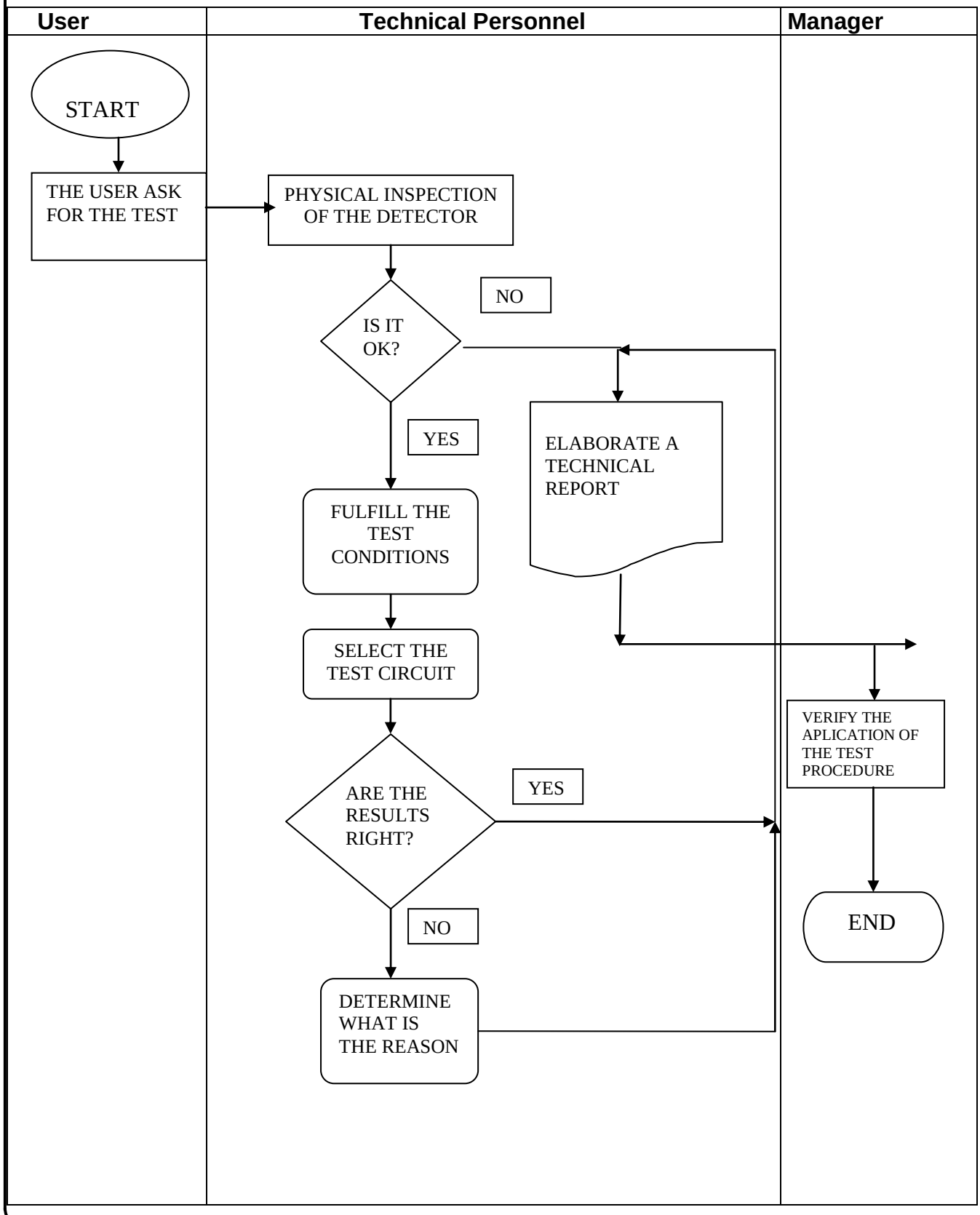
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8.- ANNEXES

Annex I

Flow Chart.


User
Technical Personnel
Manager
START
THE USER ASK FOR THE TEST
PHYSICAL INSPECTION OF THE DETECTOR
IS IT OK?
NO
YES
ELABORATE A TECHNICAL REPORT
FULFILL THE TEST CONDITIONS
SELECT THE TEST CIRCUIT
ARE THE RESULTS RIGHT?
YES
NO
DETERMINE WHAT IS THE REASON
VERIFY THE APLICATION OF THE TEST PROCEDURE
END

Annex II

Test report

TEST REPORT Nº PIN-_____

Detector number:_____

Mark:_____ Model:_____
Area:_____Window:_____

Physical Revision

Condition	YES	NO
Damage in the Body		
Integrity of the surface		
Corrosion		
Oxidation		

Instruments Employed

Instrument	Mark	Model	Serial number
Voltage Power Supply			
DMM			
Oscilloscope			
Electrometer			
Preamplifier			
Amplifier			
MCA			
Clock, Timer			
Rate Meter			
Curve tracer			

Environmental Conditions

Temperature	
Pressure (Vacuum)	

Reverse Current

Reverse voltage	
Reverse current	

Radioactive Sources

Source	Serial number	Activity	Date

Test Conditions

Test Voltage (V)	
Polarity of test voltage	
Preamplifier Pulse amplitude (± V)	
Preamplifier Rise time (ns)	
Preamplifier Decay time(μs)	
Time constant of the amplifier (μs)	
Base line restoration	
Amplifier coarse gain	
Amplifier fine gain	
Pile-up rejection	

Draw the signal obtained in the preamplifier output.	***Draw the signal obtained in the amplifier output.***

Results (as required)

Alpha Resolution (FWHM): _____ measured at 5.486 MeV (peak of Am-241 radioactive source).

X-ray Resolution (FWHM): _____ measured at 13.9 keV (peak of Am-241 radioactive source).

Diagnostic or Comments:

Tested By:	
Date:	